

Q.20 What is Data serialization for PIC programming in C? (CO3)

Q.21 Explain PIC18 timer programming in C. (CO3)

Q.22 Write the feature of PIC microcontroller. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the DC motor Interfacing with PIC18.(CO4)

Q.24 Explain the architecture of PIC Microcontroller with diagram. (CO2)

Q.25 Write short note on the following:

a) Various Data types of PIC programming in C. (CO3)

b) I/O port pins in PIC microcontroller. (CO1)

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6th Sem. / Automation and Robotics

Subject : Embedded Systems

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 There are a total of _____ ports in the PIC18F458. (CO2)

a) 3 b) 4

c) 5 d) 6

Q.2 When was the first embedded system developed?

a) 1965 b) 1990 (CO1)

c) 2000 d) 2010

Q.3 The first embedded system to be widely recognized was: (CO1)

a) Intel 4004

b) Apollo Guidance Computer

c) Shakti d) SCADA

Q.4 Microcontrollers use _____ to store data and programs. (CO2)

- a) Input/Output ports b) Memory
c) Keyboard d) Relays

Q.5 Which of the following PIC microcontroller families is known for its high-performance capabilities? (CO2)

- a) PIC 10 b) PIC 12
c) PIC 16 d) PIC 18

Q.6 The oscillator is used to generate which signal for the microcontroller? (CO3)

- a) memory b) Input/Output
c) clock d) Unknown

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 ADC stands for _____. (CO3)

Q.8 The program counter keeps track of the current _____. (CO3)

Q.9 PIC microcontrollers are widely used in industrial control applications. (True/False) (CO4)

Q.10 PIC microcontrollers are based on Complex Instruction Set Computing architecture. (True/False) (CO2)

Q.11 The brain of a microcontroller is the _____. (CO2)

Q.12 The full form of PIC in microcontroller is _____. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain DAC interfacing with PIC18? (CO3)

Q.14 Explain Data RAM allocation in C18 (CO3)

Q.15 Draw the Pin diagram of PIC18F458 microcontroller. (CO2)

Q.16 What is the function of I/O port pins/ (CO2)

Q.17 Explain various logic operations of PIC programming in C? (CO3)

Q.18 What are the essential components of an Embedded System? (CO1)

Q.19 What are relays? How are they connected with PIC18? (CO4)